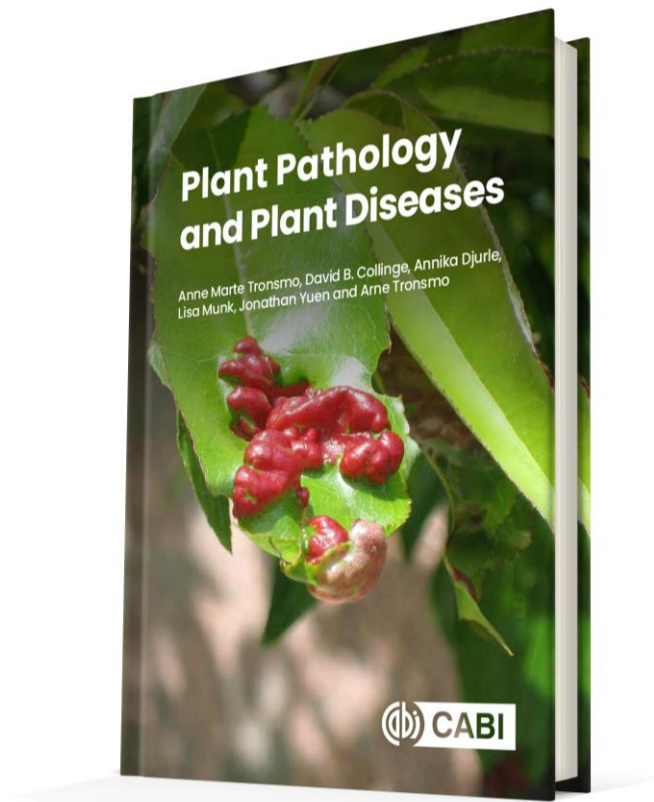




Plant Pathology and Plant Diseases

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Lisa Munk, Jonathan Yuen and Arne Tronsmo 2020

Fig. 16.1 Tractor-drawn sprayer treating the crop against potato late blight (*Phytophthora infestans*). (© Åsa Rölin.)



Fig. 16.2 Chemical structures of fungicides from major fungicide groups (Wikipedia)

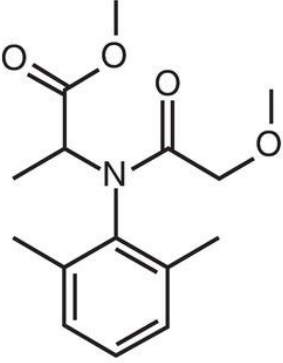
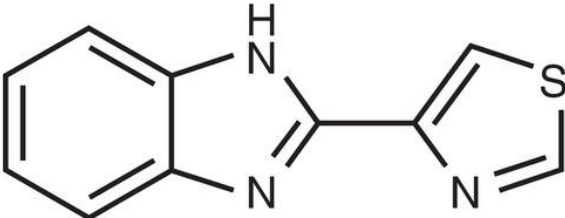
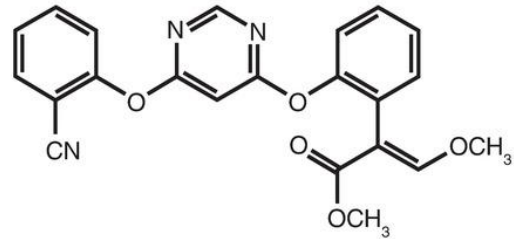
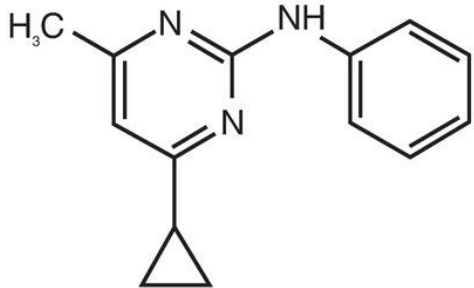
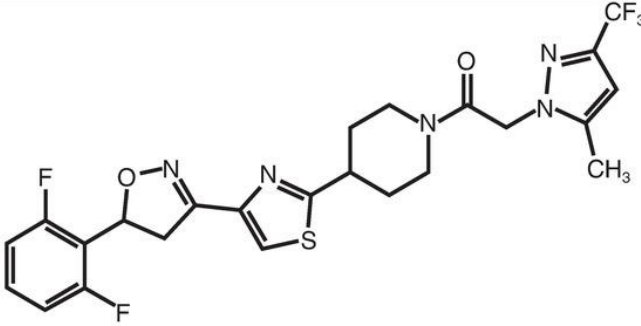
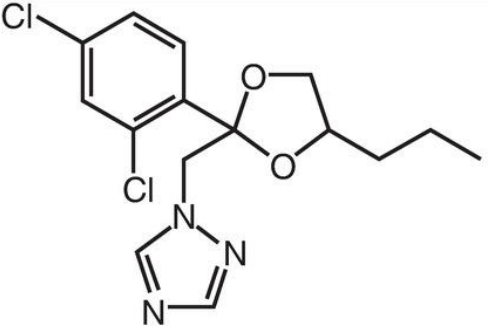
		
Group A: Metalaxyl	Group B: Thiabendazole	Group C: Azoxystrobin
		
Group D: Cyprodinil	Group F: Oxathiapiiprolin	Group G: Propiconazole

Fig. 16.3 Target cell functions for different fungicides with representative active ingredients. (© B. Andersson.)

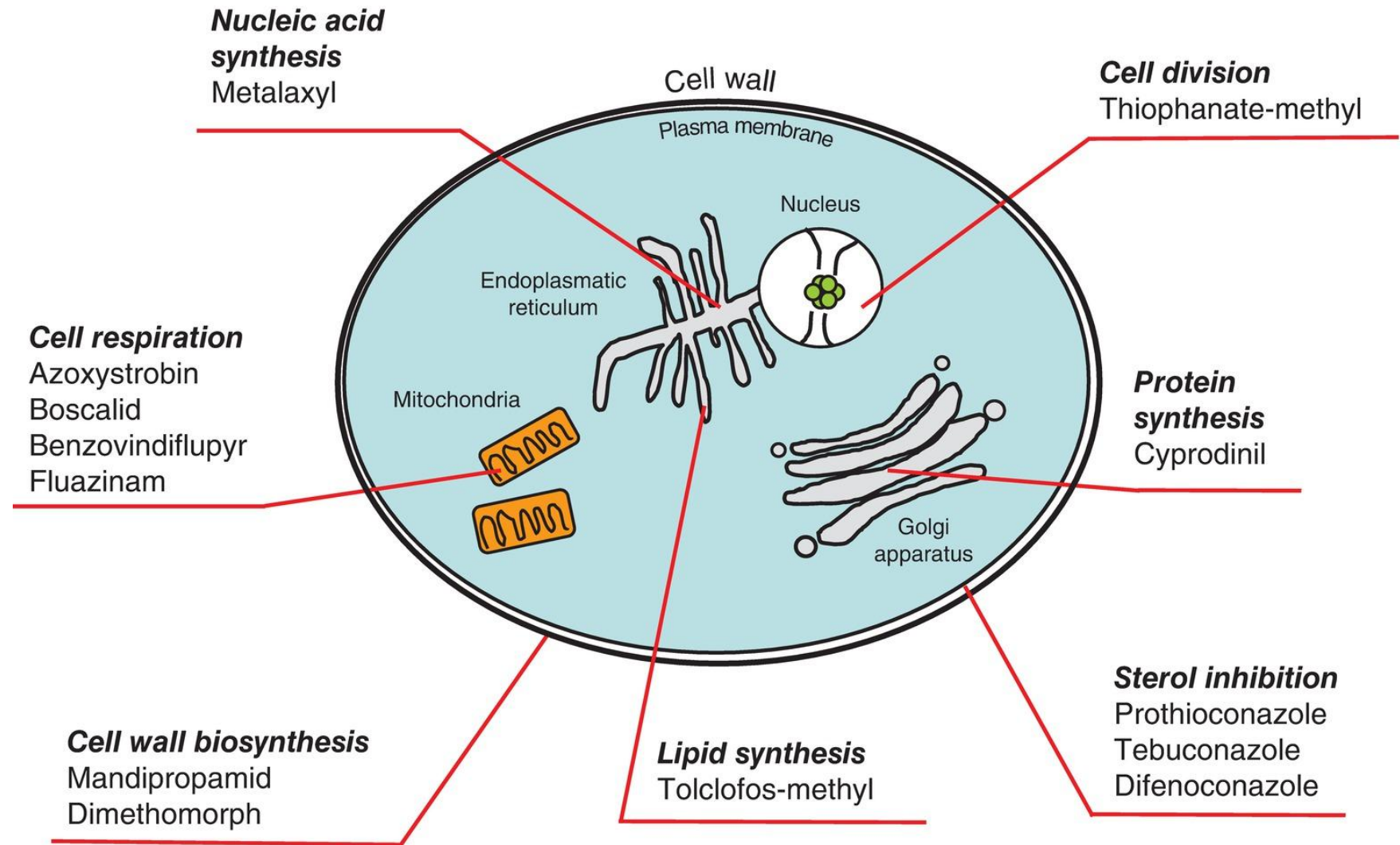


Fig. 16.4 (A) Qualitative fungicide resistance caused by a single mutation in the target fungal gene, resulting in a division of the pathogen population in one sensitive and one resistant part. (B) Quantitative fungicide resistance caused by multiple mutations in several target genes, resulting in a gradual shift in the pathogen population. (© B. Andersson.)

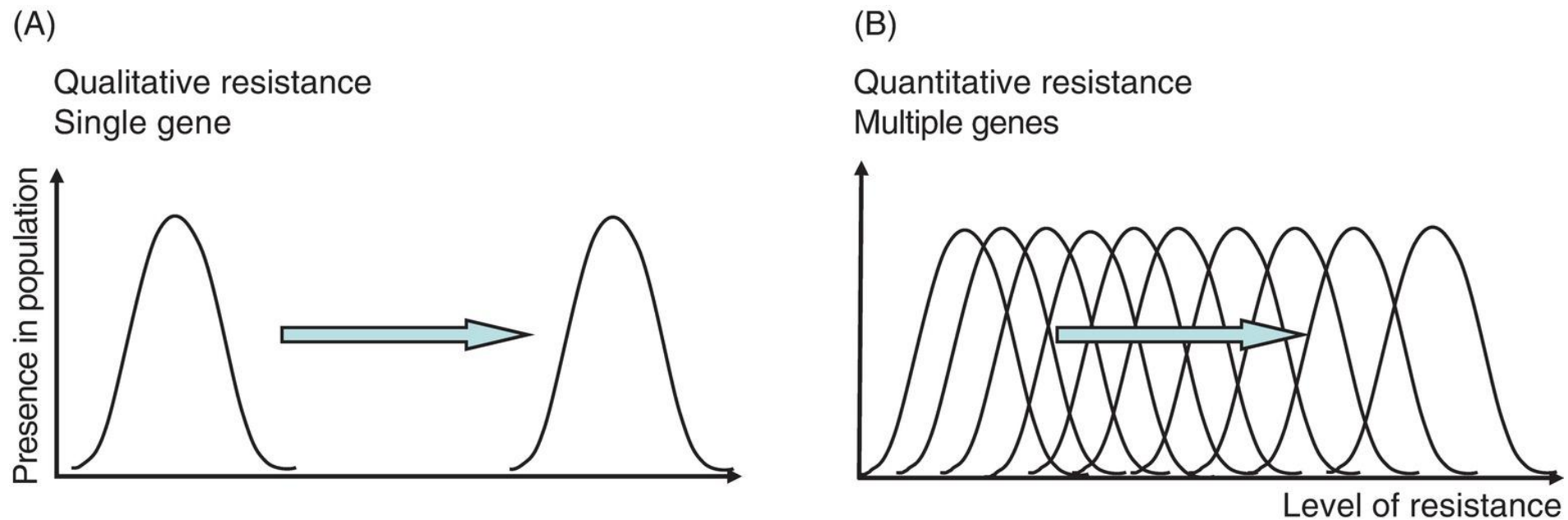


Fig. 16.5 Risk of fungicide resistance development for some combinations of fungicides and pathogens. A darker colour indicates higher risk. (© Swedish Board of Agriculture.)

HIGH • Benzimidazoles • Qols • Phenylamides			
AVERAGE • Morpholines • SDHIs • Anilino-pyrimidines			
LOW • Copper, sulphur • Chlorothalonil • Dithiocarbamates • Phthalimides			
FUNGICIDE RISK PATHOGEN RISK	LOW • <i>Rhizoctonia</i> spp. • Rust fungi • Soilborne pathogens • Smut fungi	AVERAGE • <i>Gaeumannomyces graminis</i> • <i>Zymoseptoria tritici</i> • <i>Rhynchosporium</i> spp. • <i>Pyrenophora teres</i> • <i>Ramularia collo-cygni</i>	HIGH • <i>Botrytis cinerea</i> • Powdery mildew fungi • <i>Venturia inaequalis</i> • <i>Phytophthora infestans</i>